Marco Santana

Software Engineer

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ABOUT

With over **25 years in software engineering**, I have built a career spanning full-stack development, database architecture, cloud infrastructure, and technical leadership — from enterprise platforms at Microsoft Brazil to telemedicine systems, e-procurement SaaS, and machine-learning products at H2O.ai.

Today I design and ship production software across TypeScript, Dart, and Swift. In the first quarter of 2026 alone, I shipped a multi-tenant SaaS platform, a cross-platform music notation ecosystem (iPad + macOS App Store), a community directory for a Brazilian coastal village, and a developer feedback SDK.

I value pragmatic design, clean code, and work that makes a tangible difference.

Languages: Portuguese (native) · English (full professional) · Spanish (elementary)

PROJECTS

Braxum — https://braxum.com

Multi-tenant SaaS platform with strong financial, SGA, and CRM modules. Full production system with role-based access control (5 roles), TOTP two-factor authentication, D3.js data visualization dashboards, drag-and-drop Kanban boards, vehicle inspection module with S3 uploads, and a 4-stage test pyramid. Dual-cloud Terraform infrastructure on AWS and GCP with Docker CI/CD.

Next.js 16 | TypeScript | PostgreSQL | Drizzle | D3.js | Terraform | Docker

Chordo — https://chordo.pro

Cross-platform music notation ecosystem shipped to the iPad and macOS App Store. Renders ICLF chord sheets with syllable-precise alignment, features a dual-panel visual chord editor, voicing calculator, tag-based song organization, and instrument-aware rendering. Built on a custom music theory engine with key detection, Roman numeral analysis, and transposition.

Dart | Flutter | iPad | macOS | App Store

ICLF Standard —

https://github.com/msant77/iclf-standard

Inline Chorded Lyrics Format — an open plain-text specification for notating song lyrics with precisely positioned chords. Designed for human readability so musicians can open an `.iclf` file and play instantly. Supports syllable-precise chord alignment, extensible metadata directives, and accessibility for screen readers.

Specification | CC BY 4.0

ICLF Parser —

https://github.com/msant77/iclf-parser

A Dart library that parses and validates ICLF files into structured data models. Classifies documents as valid, invalid, or recoverable — applying automatic corrections where possible. Includes a ChordMatcher for cross-ecosystem chord validation and a CLI for rendering chord sheets.

Dart | Parsing | CLI | MIT

Music Theory —

https://github.com/msant77/music-theory

A Dart package and CLI providing programmatic access to music theory — notes, intervals, chords, instruments, voicing calculation, key detection, Roman numeral analysis, transposition, and capo suggestions. Serves as the theoretical foundation for the Chordo ecosystem.

Dart | CLI | Music Theory | MIT

Ponta da Barra Info —

https://pontadabarra.info

Community directory for a coastal village in Laguna, SC, Brazil. Helps locals and tourists find service providers, transport schedules, and trails. Offline-first with bundled data, SQLite WASM for web, responsive hero carousel, and trilingual support (Portuguese, English, Spanish).

Flutter | Cloudflare Pages | SQLite WASM | Multilingual

EXPERIENCE

Independent Software Engineer

Jan 2025 – Present Santa Catarina, Brazil

Author of the projects listed above. Emphasis on offline-first architectures, internationalized interfaces, and thorough test coverage across unit, widget, integration, and end-to-end layers.

TypeScript | Dart | Next.js | Flutter | PostgreSQL | D3.js | Terraform | Docker | GitHub Actions | AWS | GCP | Cloudflare

Independent Study & Prototyping

Apr 2024 – Dec 2024

Focused period after four years at H2O.ai. Designed the ICLF chord notation specification and built the first prototypes of the Chordo ecosystem — parser library, music theory engine, and cross-platform app.

H2O.ai Frontend Engineer

May 2020 – Apr 2024 · 4 years Laguna, SC, Brazil

Built and maintained the user interface for H2O.ai's machine-learning platform in React and TypeScript. Developed shared component libraries, collaborated closely with design and backend teams, and contributed to internal developer tooling.

React | TypeScript | Storybook | Fluent UI | Go | Buf.Build | Adobe XD

BairesDev

Aug 2018 – Apr 2020 · 1 year 9 months Florianópolis, Brazil

Designed and implemented solutions for a North American e-procurement SaaS platform, progressing from senior engineer to team lead.

Technical Team Lead

Sep 2019 – Apr 2020

Led the team designing and implementing Dynamic Form Build Generation, WAND Taxonomy, and AWS Lambda/Gateway services. Onboarded new developers, provided day-to-day technical support to the team, and led grooming and planning meetings.

Senior Full Stack Engineer

Aug 2018 – Sep 2019

Designed and implemented solutions with Dynamic Form Build Generation, WAND Taxonomy, and AWS Lambda/Gateway services.

AWS Lambda | API Gateway | Node.js | React

SKILLS

LANGUAGES

TypeScript | Dart | Swift | JavaScript | C# / .NET | Go | SQL | HTML / CSS

DATA & DATABASES

PostgreSQL | SQLite | SQL Server | MongoDB | Redis | GRDB | Data Modeling

FRAMEWORKS & PLATFORMS

Next.js | React | Flutter | SwiftUI | Node.js | watchOS | D3.js | Drizzle ORM

ARCHITECTURE & PRACTICES

System Architecture | API Design | P2P / CRDTs | Offline-First | Multi-Tenancy | Technical Leadership | Test Pyramids

CLOUD & INFRASTRUCTURE

AWS | Google Cloud | Cloudflare | Terraform | Docker | GitHub Actions | Linux | NGINX

RECOMMENDATIONS

Marco is an exceptional professional with deep knowledge and remarkable technical command. He possesses the rare ability to convey complex ideas with clarity, making him an outstanding communicator and collaborator. His work is consistently of the highest quality, and his commitment to excellence is evident in everything he does. I wholeheartedly recommend Marco for any team looking for a talented, dedicated, and thoughtful engineer.

Claudio Fernando Maciel

CTO, Brokerme

I had the pleasure of working with Marco at Microsoft Brazil. He is a focused, hard-working professional with deep expertise across databases, web development, and software architecture. Marco is always eager to learn and share knowledge, and he brings a rare combination of technical skill and genuine teamwork. Any organization would benefit from his contributions.

Rob Ranches

BI Architect / Data Scientist, Bausch Health

Marco is an extraordinary professional with solid expertise in software development and architecture. He approaches every project with diligence, creativity, and a genuine desire to deliver the best possible solution. His ability to work across technologies and lead teams makes him an invaluable asset. I highly recommend him.

Jean Paulo da Silva Ramos

CEO, Ecovilas

EDUCATION

FECAP — Fundação Escola de Comércio Álvares Penteado

Data Processing Technology

1993 – 1995

CERTIFICATIONS

Designing Desktop Applications with Microsoft Visual Basic 6.0

Exam 70-176

Designing Distributed Applications with Microsoft Visual Basic 6.0

Exam 70-175

Collaborative Solutions with MS Outlook 2000 / Exchange Server 5.5

Exam 70-105